Project: Building a Healthcare AI Assistant

This Project provides hands-on experience in building an end-to-end healthcare AI assistant using Microsoft Foundry. The solution is designed to support hospital patient-services teams by answering routine questions, retrieving appointment information, processing discharge documents, and generating safe, grounded responses. It combines generative AI, RAG, function calling, document understanding, multi-agent orchestration, and MCP integration within a single practical workflow.



The main concepts covered in this lab include:

AI Agent Development: Create a healthcare assistant with conversation memory, clear instructions, and safety rules.

Function Calling: Connect the agent to synthetic patient and appointment data for record lookup and appointment management.

Retrieval-Augmented Generation: Index hospital policy documents and use file search to generate evidence-based answers.

Document Processing: Extract information from discharge-note PDFs and convert it into Markdown and structured JSON.

Multi-Agent and MCP Integration: Coordinate specialist agents and connect to external documentation tools through the Model Context Protocol.

In future healthcare applications, this solution can be extended to integrate with real hospital systems, patient portals, FHIR APIs (Fast Healthcare Interoperability Resources APIs), billing platforms, and clinical document repositories. It can help reduce repetitive workloads, improve access to information, support faster patient coordination, and allow healthcare staff to focus on cases that require human judgment and care.